

Asynchronous Activity 1

Student Early Warning Tool Using KNIME

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Course / Section	CS0065 Intelligent Systems / AN43
Repository	Fernando-Ken-Dylen-CS0065-AN43
Project folder	Technical-Assessment-01/DemoEarlyWarningTool

1. Completed KNIME workflow and model result

The screenshot displays the KNIME Analytics Platform interface. The top bar shows the 'Home' button, 'Activity1' tab, and navigation icons. The left sidebar contains 'Info', 'Nodes', 'Explorer', 'K-AI', and 'Monitor' sections. The main workspace shows a workflow starting with a 'CSV Reader' node, followed by 'Column Filter', 'Normalizer', and 'Table Partitioner'. The workflow then branches into three parallel paths: 'Decision Tree Learner', 'Logistic Regression Learner', and 'Random Forest Learner'. Each path leads to a corresponding predictor node ('Decision Tree Predictor', 'Logistic Regression Predictor', and 'Random Forest Predictor') and finally to a 'Scorer' node. The 'Scorer' node is highlighted, and its configuration panel on the right shows 'First column' as 'risk_status' and 'Second column' as 'Prediction (risk_status)'. The 'Sorting strategy' is set to 'Insertion order'. The 'Missing values' section has 'Ignore' and 'Fail' buttons. Below the workflow, the 'Scorer' node's output is displayed as a table with 2 rows and 2 columns. The table shows the results of the classification for two records.

Scorer

Compares two columns by their attribute value pairs and shows the confusion matrix, i.e. how many rows of which attribute and their classification match. Additionally, it is possible to highlight cells of this matrix to determine the underlying rows. The dialog allows you to select two columns for comparison; the values from the first selected column are represented in the confusion matrix's rows and the values from the second column by the confusion matrix's columns. The output of the node is the confusion matrix with the number of matches in each cell. Additionally, the second out-port reports a number of accuracy statistics such as True-Positives, False-Positives, True-Negatives, False-Negatives, Recall, Precision, Sensitivity, Specificity, F-measure, as well as the overall accuracy and Cohen's kappa.

External resources

No links have been added yet

Input ports

- Type: Table
- Name: Input table
- Table containing the columns to compare.

Output ports

- Type: Table
- Name: Confusion matrix
- The confusion matrix.
- Type: Table
- Name: Accuracy statistics
- The accuracy statistics table.

Extension

KNIME Base nodes
provided by KNIME AG, Zurich, Switzerland

Scorer

First column: risk_status

Second column: Prediction (risk_status)

Sorting strategy: Insertion order

Reverse order: ☐

Use name prefix: ☐

Missing values: Ignore Fail

Table

#	RowID	Not At Risk	At Risk
1	Not At Risk	5	0
2	At Risk	0	4

The workflow reads and prepares the student dataset, partitions it, and compares Decision Tree, Logistic Regression, and Random Forest models. The visible Scorer result shows 5 Not At Risk and 4 At Risk records classified correctly.

2. Required Demo project files

DemoEarlyWarningTool - required deliverables

D:\github\Fernando-Ken-Dylen-CS0065-AN43\Technical-Assessment-01\DemoEarlyWarningTool

Name	Type	Size
DemoEarlyWarningTool.knwf	KNIME workflow	126.0 KB
README.md	Markdown document	2.2 KB
student_performance_knime.csv	CSV dataset	0.9 KB

Prepared locally; Git commit and public upload are pending user confirmation.

The DemoEarlyWarningTool folder contains the exported .knwf workflow, the required README.md file, and the dataset used by the CSV Reader. The final evidence PDF is added after public-repository verification.

3. Repository organization

Repository layout prepared for Asynchronous Activity 1

D:\github\Fernando-Ken-Dylen-CS0065-AN43

```
Fernando-Ken-Dylen-CS0065-AN43/
|-- .gitignore
|-- README.md
`-- Technical-Assessment-01/
    |-- README.md
    |-- DemoEarlyWarningTool/
        |-- DemoEarlyWarningTool.knwf
        |-- README.md
        |-- student_performance_knime.csv
        |-- Fernando_Ken_Dylen_KNIME_GitHub_Evidence.pdf (to be finalized)
    |-- Python and Streamlit implementation files

Git status: local changes prepared; no commit or push performed yet.
```

The activity is organized under Technical-Assessment-01 while preserving the required DemoEarlyWarningTool project folder and its deliverables.

4. Public GitHub repository

The screenshot shows a public GitHub repository for user **krfernando-FEUA**, named **Fernando-Ken-Dylen-CS0065-AN43**. The repository is public and has 0 forks and 0 stars. The main branch is **main**. The repository structure is as follows:

- Technical-Assessment-01
 - DemoEarlyWarningTool
 - DemoEarlyWarningTool.knwf
 - Fernando_Ken_Dylen_KNIME_...
 - README.md
 - student_performance_knime.csv

The **DemoEarlyWarningTool** folder is selected, showing a list of files and their commit history. The files are:

Name	Last commit message	Last commit date
..		
DemoEarlyWarningTool.knwf	Add Technical Assessment 01 KNIME project	1 minute ago
Fernando_Ken_Dylen_KNIME_GitHub_Evidence....	Add Technical Assessment 01 KNIME project	1 minute ago
README.md	Add Technical Assessment 01 KNIME project	1 minute ago
student_performance_knime.csv	Add Technical Assessment 01 KNIME project	1 minute ago

The **README.md** file is selected, showing the title **Asynchronous Activity 1 Student Early Warning Tool Using KNIME**.

The public GitHub repository displays the required DemoEarlyWarningTool folder and files after a successful commit and push.

5. Public-access verification

The screenshot shows the GitHub interface for a public repository. The repository name is `krfernando-FEUA / Fernando-Ken-Dylen-CS0065-AN43`. The left sidebar shows the file structure with the following files and folders:

- Technical-Assessment-01
 - DemoEarlyWarningTool
 - DemoEarlyWarningTool.knwf
 - Fernando_Ken_Dylen_KNIME_...
 - README.md
 - student_performance_knime.csv
- README.md
- app.py
- confusion_matrices.json
- model_comparison.csv
- model_metadata.json
- requirements.txt

The main content area shows the commit history for the `DemoEarlyWarningTool` directory. The commit message is "Add Technical Assessment 01 KNIME project" by `krfernando-FEUA`. The commit hash is `01c25b4` and it was made 1 minute ago. The commit history table is as follows:

Name	Last commit message	Last commit date
..		
DemoEarlyWarningTool.knwf	Add Technical Assessment 01 KNIME project	1 minute ago
Fernando_Ken_Dylen_KNIME_GitHub_Evidence....	Add Technical Assessment 01 KNIME project	1 minute ago
README.md	Add Technical Assessment 01 KNIME project	1 minute ago
student_performance_knime.csv	Add Technical Assessment 01 KNIME project	1 minute ago

Below the commit history, the `README.md` file is displayed, showing the title "Asynchronous Activity 1 Student Early Warning Tool Using KNIME".

The repository opens without an authenticated GitHub session, confirming that its visibility is Public and that the files can be reviewed.

Verification checklist

- Repository link opens and visibility is Public.
- DemoEarlyWarningTool folder is visible.
- DemoEarlyWarningTool.knwf uses the .knwf extension.
- README.md is present and documents the project, models, and run instructions.
- student_performance_knime.csv is included and can be selected in CSV Reader.
- Evidence PDF opens from the GitHub repository.
- Workflow archive contains KNIME workflow metadata and node settings.